



```
In [2]: import pandas as pd
```

```
In [9]: df = pd.read_csv("SampleSuperstore.csv")
```

```
In [15]: df.head() # for a data analyst its a First Quality Check
```

```
Out[15]:
```

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category
0	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
1	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
2	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies
3	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture
4	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies

```
In [21]: df.shape
```

```
Out[21]: (9994, 13)
```

```
In [23]: df.columns
```

```
Out[23]: Index(['Ship Mode', 'Segment', 'Country', 'City', 'State', 'Postal Code',  
               'Region', 'Category', 'Sub-Category', 'Sales', 'Quantity', 'Discount',  
               'Profit'],  
              dtype='object')
```

```
In [25]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Ship Mode       9994 non-null   object
1   Segment         9994 non-null   object
2   Country         9994 non-null   object
3   City            9994 non-null   object
4   State           9994 non-null   object
5   Postal Code     9994 non-null   int64
6   Region          9994 non-null   object
7   Category        9994 non-null   object
8   Sub-Category    9994 non-null   object
9   Sales           9994 non-null   float64
10  Quantity        9994 non-null   int64
11  Discount        9994 non-null   float64
12  Profit          9994 non-null   float64
dtypes: float64(3), int64(2), object(8)
memory usage: 1015.1+ KB

```

```
In [27]: df.isnull().sum()
```

```

Out[27]: Ship Mode      0
        Segment      0
        Country      0
        City         0
        State        0
        Postal Code   0
        Region       0
        Category     0
        Sub-Category  0
        Sales        0
        Quantity     0
        Discount     0
        Profit       0
        dtype: int64

```

```
In [33]: df.duplicated().sum()
```

```
Out[33]: 17
```

```
In [35]: df[df.duplicated()]
```

Out[35]:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Cat
950	Standard Class	Home Office	United States	Philadelphia	Pennsylvania	19120	East	St
3406	Standard Class	Home Office	United States	Columbus	Ohio	43229	East	Fu
3670	Standard Class	Consumer	United States	Salem	Oregon	97301	West	St
4117	Standard Class	Consumer	United States	Los Angeles	California	90036	West	St
4553	Standard Class	Consumer	United States	San Francisco	California	94122	West	St
5905	Same Day	Home Office	United States	San Francisco	California	94122	West	St
6146	Standard Class	Corporate	United States	San Francisco	California	94122	West	St
6334	Standard Class	Consumer	United States	New York City	New York	10011	East	St
6357	Standard Class	Corporate	United States	Seattle	Washington	98103	West	St
7608	Standard Class	Consumer	United States	San Francisco	California	94122	West	St
7735	Standard Class	Corporate	United States	Seattle	Washington	98105	West	St
7759	Standard Class	Corporate	United States	Houston	Texas	77041	Central	St
8032	First Class	Consumer	United States	Houston	Texas	77041	Central	St
8095	Second Class	Consumer	United States	Seattle	Washington	98115	West	St
9262	Standard Class	Consumer	United States	Detroit	Michigan	48227	Central	Fu
9363	Standard Class	Home Office	United States	Seattle	Washington	98105	West	Fu
9477	Second Class	Corporate	United States	Chicago	Illinois	60653	Central	St

```
In [37]: df = df.drop_duplicates()
```

```
In [41]: df.shape
```

Out[41]: (9977, 13)

```
In [43]: df.duplicated().sum()
```

```
Out[43]: 0
```

```
In [51]: df["Segment"].unique()
```

```
Out[51]: array(['Consumer', 'Corporate', 'Home Office'], dtype=object)
```

```
In [53]: df["Category"].unique()
```

```
Out[53]: array(['Furniture', 'Office Supplies', 'Technology'], dtype=object)
```

```
In [58]: df["Region"].unique()
```

```
Out[58]: array(['South', 'West', 'Central', 'East'], dtype=object)
```

```
In [60]: df["Sub-Category"].unique()
```

```
Out[60]: array(['Bookcases', 'Chairs', 'Labels', 'Tables', 'Storage',  
                'Furnishings', 'Art', 'Phones', 'Binders', 'Appliances', 'Paper',  
                'Accessories', 'Envelopes', 'Fasteners', 'Supplies', 'Machines',  
                'Copiers'], dtype=object)
```

```
In [62]: df.to_csv("SuperstoreSales_Cleaned.csv", index = False)
```

```
In [ ]:
```



```
In [26]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [28]: df = pd.read_csv("SuperstoreSales_Cleaned.csv")
```

```
In [30]: df.head()
```

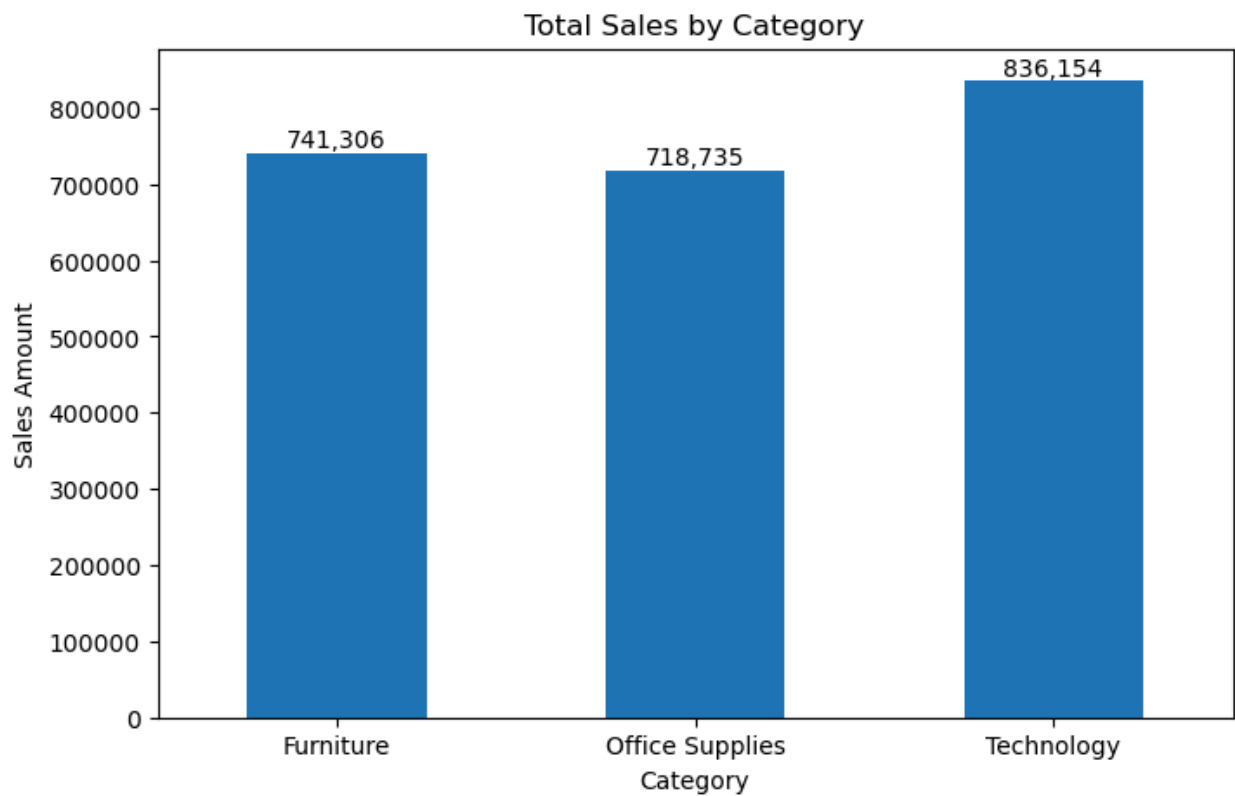
```
Out[30]:
```

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category
0	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
1	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture
2	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies
3	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture
4	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies

```
In [32]: category_sales = df.groupby('Category')['Sales'].sum()
ax = category_sales.plot(kind = 'bar', figsize=(8,5))
plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales Amount")
plt.xticks(rotation = 0)

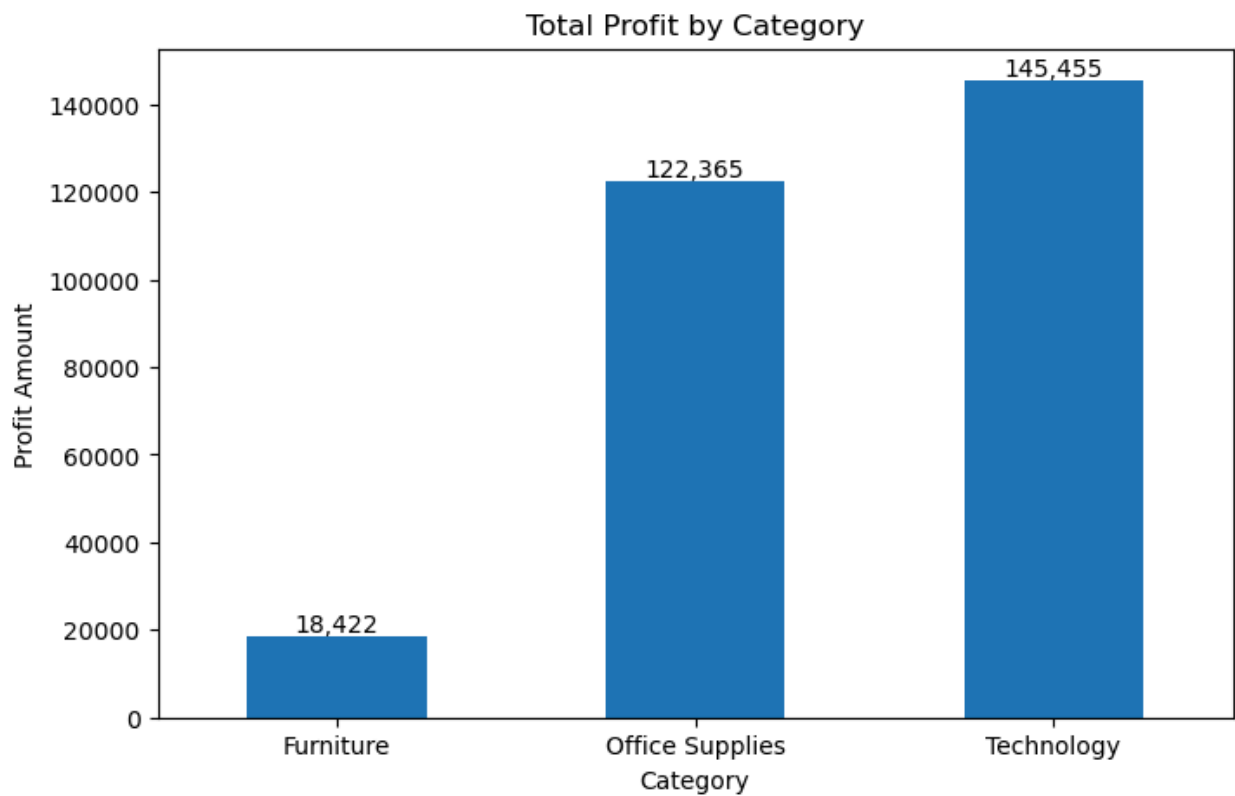
for i, value in enumerate(category_sales):
    ax.text(i,value , f'{value:,.0f}', ha = 'center', va= 'bottom')

plt.show()
```



```
In [34]: category_profit = df.groupby('Category')['Profit'].sum()
category_profit.plot(kind = 'bar', figsize = (8,5))
plt.title("Total Profit by Category")
plt.xlabel('Category')
plt.ylabel('Profit Amount')
plt.xticks(rotation = 0)

for i, value in enumerate(category_profit):
    plt.text(i,value, f'{value:,.0f}', ha = 'center', va = 'bottom')
plt.show()
```



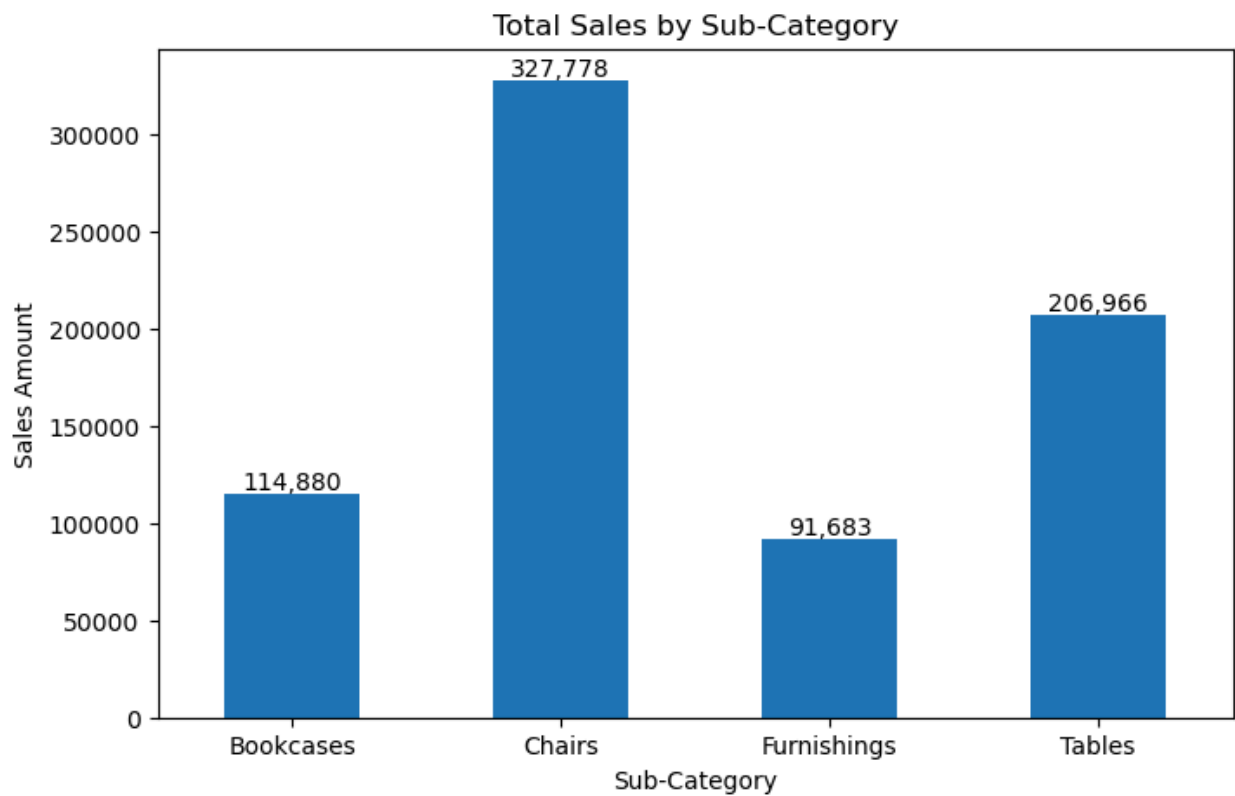
```
In [36]: furniture_df = df[df['Category'] == 'Furniture']
```

```
In [38]: Subcategory_Sales = furniture_df.groupby('Sub-Category')['Sales'].sum()

Subcategory_Sales.plot(kind = 'bar', figsize = (8,5))
plt.title('Total Sales by Sub-Category')
plt.xlabel('Sub-Category')
plt.ylabel('Sales Amount')
plt.xticks(rotation = 0)

for i, value in enumerate(Subcategory_Sales):
    plt.text(i, value, f'{value:,.0f}', ha = 'center' ,va = 'bottom')

plt.show()
```

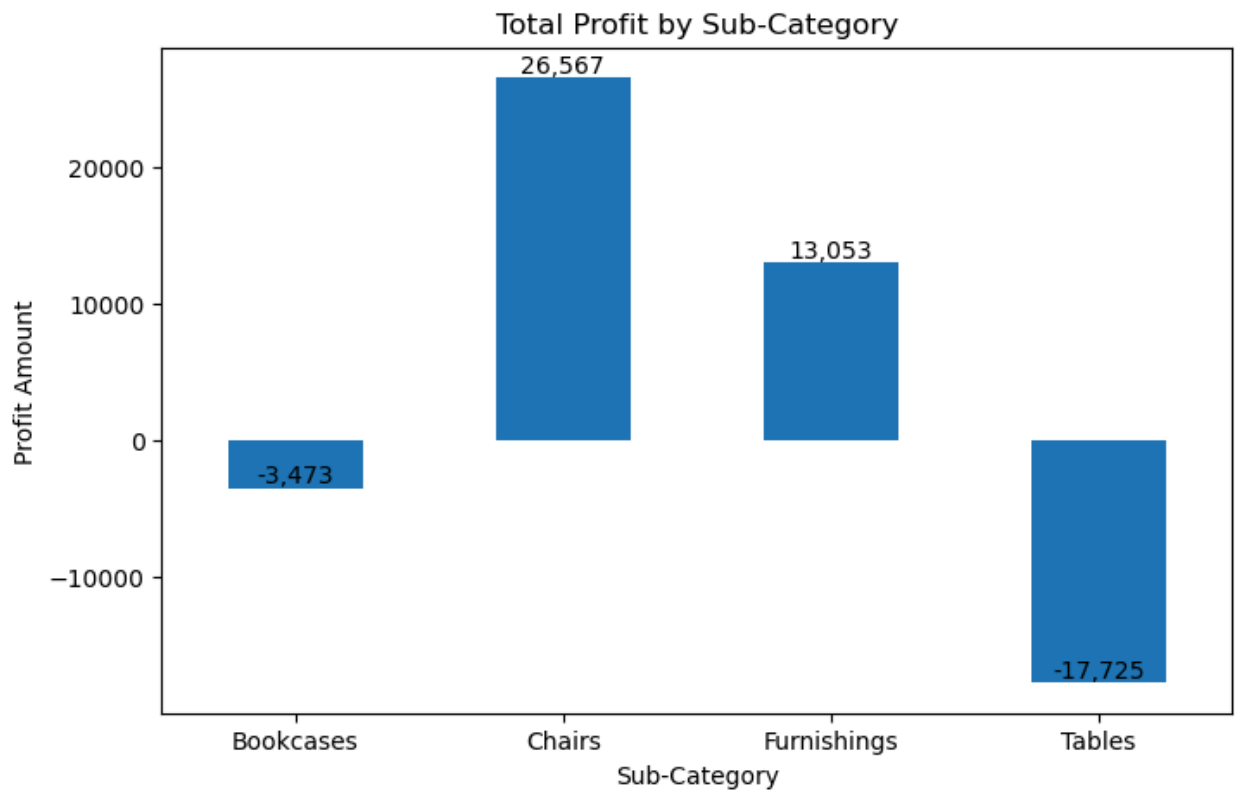


```
In [40]: Subcategory_Profit = furniture_df.groupby('Sub-Category')['Profit'].sum()

Subcategory_Profit.plot(kind = 'bar', figsize = (8,5))
plt.title('Total Profit by Sub-Category')
plt.xlabel('Sub-Category')
plt.ylabel('Profit Amount')
plt.xticks(rotation = 0)

for i, value in enumerate(Subcategory_Profit):
    plt.text(i, value, f'{value:,.0f}', ha = 'center' ,va = 'bottom')

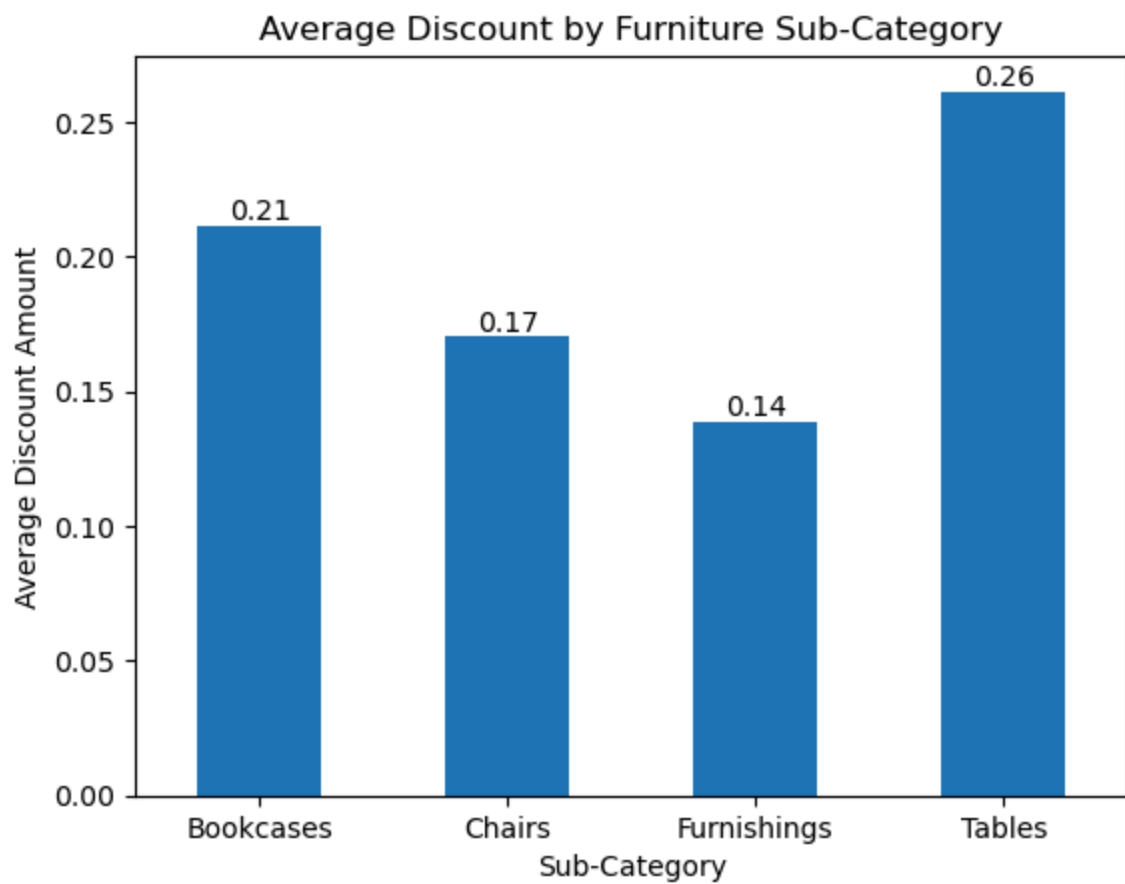
plt.show()
```

```
In [42]: Subcategory_AvgDiscount = furniture_df.groupby('Sub-Category')['Discount'].mean()
Subcategory_AvgDiscount.plot(kind = "bar")

plt.title('Average Discount by Furniture Sub-Category')
plt.xlabel('Sub-Category')
plt.ylabel('Average Discount Amount')
plt.xticks(rotation = 0)

for i, value in enumerate(Subcategory_AvgDiscount):
    plt.text(i, value, f'{value:,.2f}', ha='center', va='bottom')
plt.show()
```



```
In [52]: tables_df = df[df['Sub-Category'] == 'Tables']
```

```
In [54]: tables_df
```

Out[54]:

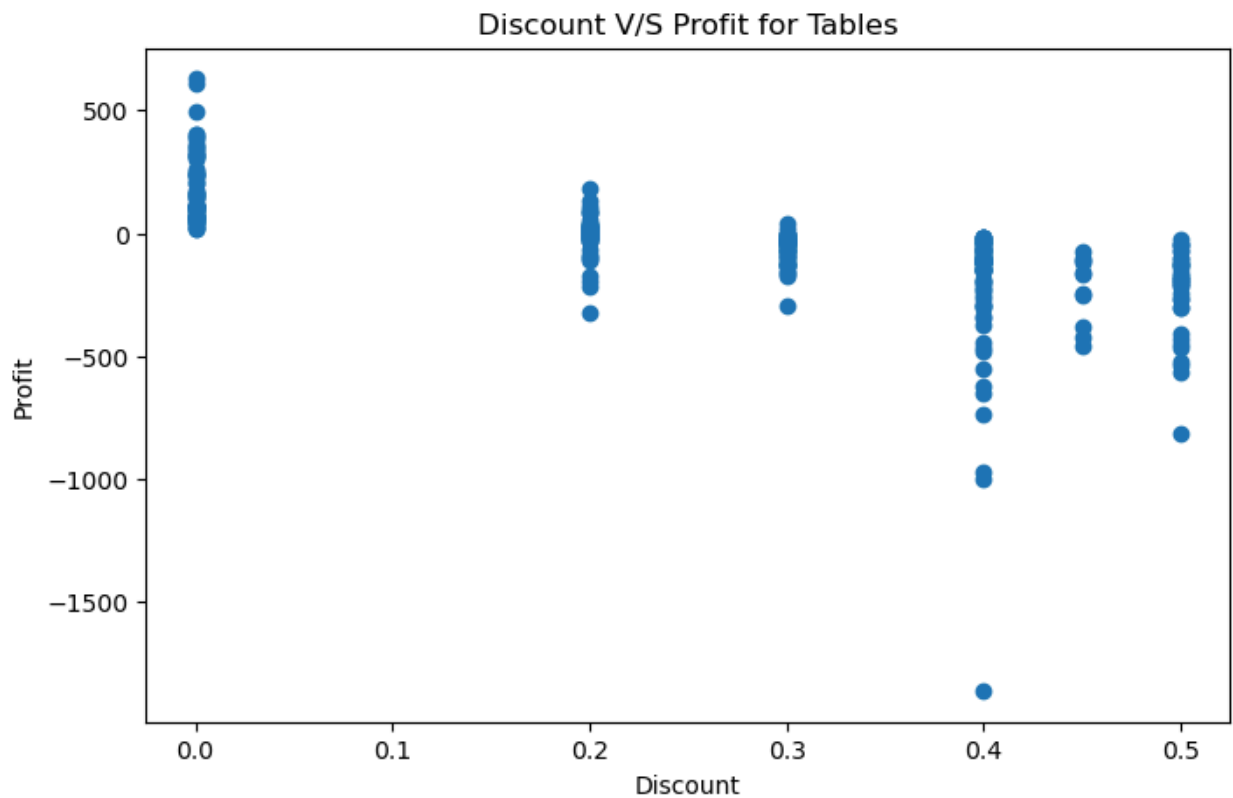
	Ship Mode	Segment	Country	City	State	Postal Code	Region	Cate
3	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Fur
10	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Fur
24	Standard Class	Consumer	United States	Orem	Utah	84057	West	Fur
117	Standard Class	Consumer	United States	Seattle	Washington	98103	West	Fur
125	Standard Class	Consumer	United States	Bloomington	Illinois	61701	Central	Fur
...	...	...	...	...	...	...	...	...
9840	Standard Class	Corporate	United States	Seattle	Washington	98103	West	Fur
9882	Standard Class	Home Office	United States	Springfield	Ohio	45503	East	Fur
9920	Second Class	Corporate	United States	Los Angeles	California	90049	West	Fur
9938	Second Class	Corporate	United States	Los Angeles	California	90045	West	Fur
9963	Second Class	Consumer	United States	Lafayette	Louisiana	70506	South	Fur

319 rows × 13 columns

```
In [60]: tables_df.shape[0]
```

Out[60]: 319

```
In [76]: plt.figure(figsize = (8,5))
plt.scatter(tables_df['Discount'] , tables_df['Profit'])
plt.xlabel('Discount')
plt.ylabel('Profit')
plt.title('Discount V/S Profit for Tables')
plt.show()
```



```
In [78]: tables_df[["Discount" , 'Profit']].corr()
```

```
Out[78]:
```

	Discount	Profit
Discount	1.00000	-0.67177
Profit	-0.67177	1.00000

```
In [80]: tables_df.groupby('Region')['Profit'].mean()
```

```
Out[80]: Region
Central    -49.439589
East       -137.817251
South      -90.648194
West        12.781097
Name: Profit, dtype: float64
```

```
In [82]: tables_df.groupby('Segment')['Profit'].mean()
```

```
Out[82]: Segment
Consumer    -61.182628
Corporate   -46.287723
Home Office -57.239717
Name: Profit, dtype: float64
```

```
In [84]: tables_df.groupby('Ship Mode')['Profit'].mean()
```

```
Out[84]: Ship Mode
         First Class      -29.050351
         Same Day        -53.782024
         Second Class    -54.437375
         Standard Class  -62.684275
         Name: Profit, dtype: float64
```

```
In [ ]:
```